

Quick Guide



Revision C (05/2020)

en-UK

DISCLAIMER

This document is intended to provide an overview of LIAISON® XS most frequently used functionalities from an instrument perspective. It is not intended to provide a complete overview about all functionalities, safety measures, precautions, cautions and restrictions for the analyzer, which are covered in the LIAISON® XS User Manual, and for LIAISON® XS reagents and consumables, which are covered in the related Instructions For Use.

This document refers to access at the highest level.

Working with lower level access may cause a lack of some functionalities.

LIAISON® XS may only be used by authorized and trained personnel. This guide cannot be considered as a substitute of official trainings supported by DiaSorin.

In the event of problems or doubts about using the system, contact the local support.



DiaSorin S.p.A.

Strada per Crescentino - 13040 Saluggia (Vercelli) - Italy

www.diasorin.com



AMENDMENT INDEX

Revision	Impacted chapter/section	Description of Change(s)
C	-	Grammatical corrections
C	1.1	System overview description update
C	1.4	Flaps and drawers description update
C	2	Update of Glossary to include System Liquid
C	5.7	Inclusion of System Liquid in Needs page
C	7.1	Inclusion of System Liquid in Status-Summary page
C	7.2	Status Summary page image update
C	7.4	Inclusion of System Liquid loading handling
C	8.2	Inclusion of time based filter for backup creation

Table of Contents

1	Instrument scheme	1-1
1.1	System Overview	1-1
1.2	System Modules Left.....	1-2
1.3	System Modules Right	1-3
1.4	Flaps and Drawer	1-4
2	Glossary.....	2-1
3	Switch on - off	3-1
3.1	Switching the system on.....	3-1
3.2	Switching the system off	3-3
4	Calibration	4-1
4.1	Ancillary loading.....	4-1
4.2	Integral loading	4-2
5	Test performance.....	5-1
5.1	Sample Loading	5-1
5.2	Assay assignment for patient sample	5-3
5.3	Assay assignment for control samples.....	5-6
5.4	Assay assignment for external calibrators	5-8
5.5	Integrals loading	5-9
5.6	Ancillary loading.....	5-9
5.7	Upfront estimation.....	5-10
5.8	Routine start.....	5-11
6	Results	6-1

7	Consumables & Temperatures	7-1
7.1	Consumables value check.....	7-1
7.2	Temperature value check.....	7-3
7.3	Cuvette & Tip Loading.....	7-4
7.4	Other consumables loading	7-7
8	Maintenance & back-up	8-1
8.1	Maintenance	8-1
8.2	Back-up.....	8-2
9	Assay definition	9-1
10	Control definition	10-1
11	Events	11-1
12	Left Header	12-1

Annex 1: FLAGS

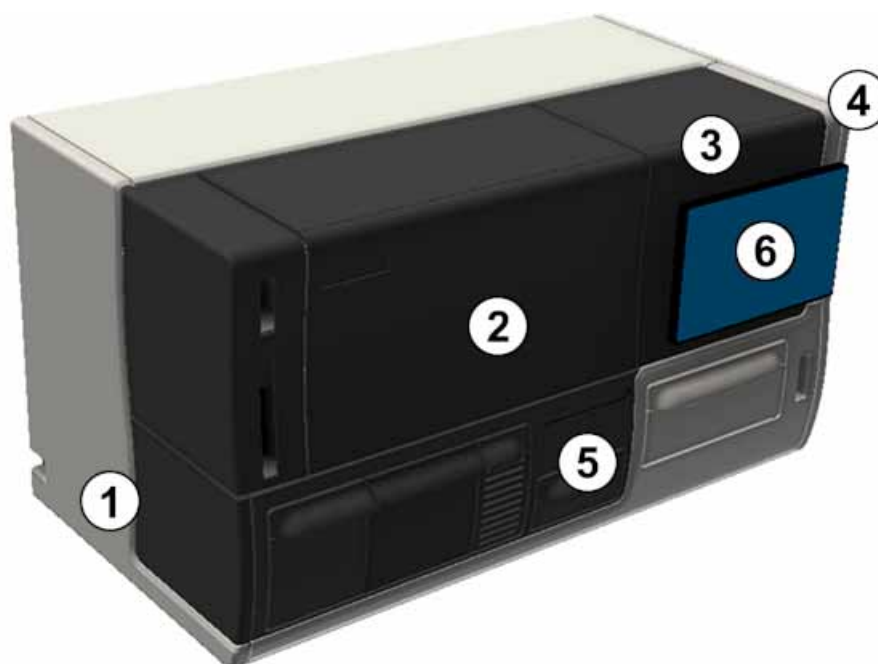
Annex 2: INSTRUMENT SCHEME (system modules)

Annex 3: INTEGRAL SYMBOLS

1 Instrument scheme

(See also User Manual, chapter “System Description”)

1.1 System Overview



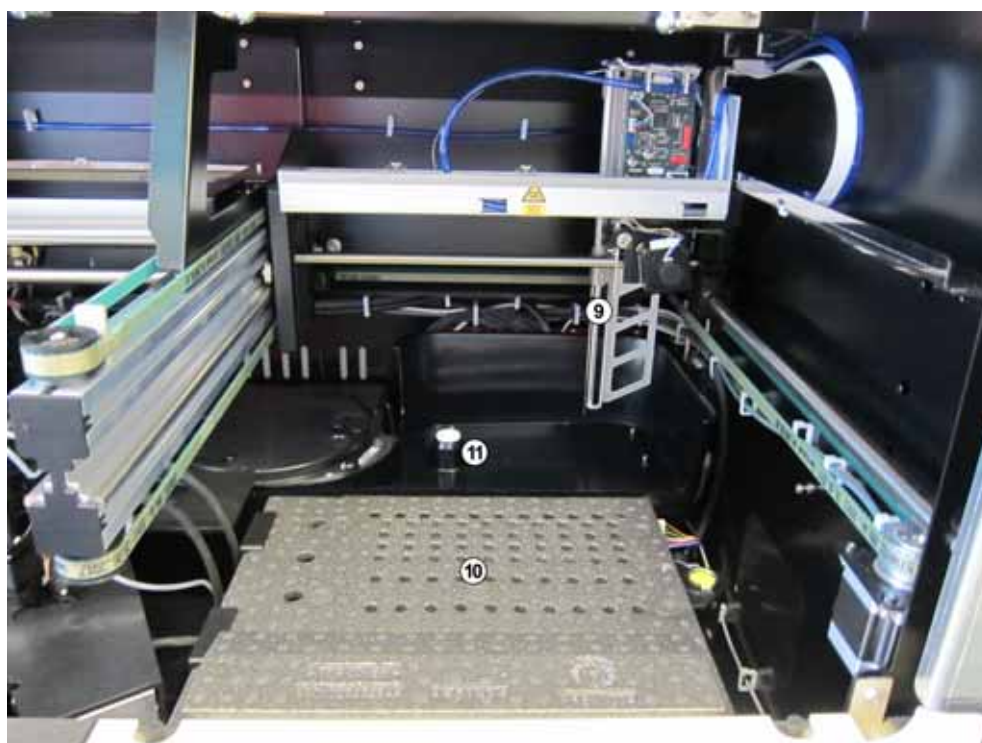
No.	Description
1)	Left side cover
2)	Left top cover
3)	Right top cover
4)	Right side cover
5)	Front Module (starter reagents, drawer for disposable cuvette and tip trays, loading bays for samples, solid waste bin, reagents area and resuspension tool)
6)	Panel-PC with touch screen

1.2 System Modules Left



No.	Description
1)	Air pipettor with disposable tip adapter for sample pipetting
2)	Cuvette gripper
3)	Cuvette trays area
4)	Disposable tips area
5)	Samples area
6)	Washer
7)	Incubator
8)	Reader

1.3 System Modules Right



No.	Description
9)	Pipettor with steel needle (SPOLV) for reagent pipetting
10)	Reagents area
11)	Pipettor wash station

1.4 Flaps and Drawer



No.	Description
1)	Top Cover Right
2)	Top Cover Left
3)	Panel-PC with touch screen
4)	Starters reagent flap
5)	Disposal tips and cuvettes drawer
6)	Sample bay flap
7)	Solid waste Bin
8)	Status Light
9)	Reagent module flap
10)	Integrated re-suspension tool

2 Glossary

Ancillary (reagents)	Anything that can be introduced by operator into Ancillary Area rack, including external reagents for some kits and Light Check
Barcode Reader	Assembly to read the sample barcode
BGW	Test to check quality of instrument washing
Control Rack	8 positions module to host some types of control bottles and external calibrators
Cuvette	Single cavity plastic module, in which immunometrical reaction can take place
CV%	Statistical variable that shows dispersion rate of measurements
EASY Waste	Carton disposable box to increase safety during waste disposal.
Gripper	Mechanism to handle cuvettes loading from trays, exchange cuvettes between incubator and washer and move cuvettes from washer to reader
Incubator	The incubator offers 32 cuvette positions in the outer ring and 25 cuvette positions in the inner ring in which cuvettes are incubated and pipetted.
Instrument	Part of LIAISON[®] XS system that includes only the LIAISON[®] XS instrument and tanks, but printer and connection cables
Integral	Reagent cartridge (made up of vials) to be inserted into the Reagent Area
Integral Holder	Holder used for the handling of integrals outside the instrument. Up to 16 integrals can be transported together.
Kit	Set of reagents used to carry out a specific assay; it may consist of one or more integrals, external calibrators and ancillary reagents.
LC-le	Test to verify precision of the left dispensing pipettor, carried out by Light Check solution
LC-ri	Test to verify precision of the right dispensing pipettor, carried out by Light Check solution
Light Check	Test tool provided as lyophilised material
Light Check adaptor	Plastic adaptor used to insert the Light Check bottles into the Ancillary Area rack
Liquid Waste Basin	Basin used to collect spillage of contaminated liquid coming from broken liquid waste tank
Liquid Waste Tank	Tank to collect all the liquid waste of the instrument
Prime	Start-up cycle for individual parts of instrument involved in fluidics, carried out by using System Liquid, DI Water, Wash Buffer, Starter Reagents and Cleaning Solution

2. Glossary

Rack Holder	12 positions holder used for the handling of sample and control racks outside the instrument.
Reader	Reading area in which chemical reaction and measurements occur
Reagent Area	Integral and ancillary loading area
RF-Tag	Micro-chip present on integrals, starters and ancillaries to allow recognition and data storage
RLU	Relative Light Unit (signal measurement unit)
Sample Area	Loading area for samples
Sample Rack	12 positions module to host sample tubes
Samples	Anything that can be introduced by operator into Sample Area racks, including patient samples, controls and external calibrators
Solid Waste Bin	Container of the system used to host EASY Waste
SPOLV	Right arm non magnetic stainless steel probe
Starter (reagents)	Reagents dispensed during the reading to generate chemiluminescent signal
System	The complete structure installed in the laboratory
System Liquid	Solution present in the fluidic line of the pipettor, used to wash the pipettor probe
Tip	Single cavity plastic probe used to aspirate samples with left pipettor arm
Tray	96 positions modules for housing Disposable Tips and 35 positions modules for housing cuvettes equipped in the Disposable Tip and Cuvette drawer
Vial	Container for just one reagent; more vials form an integral
Wash Buffer	Solution (to be diluted 1:10 in distilled H ₂ O) used to wash cuvettes
Wash Station	Washing well for right pipettor needle
Washer	Washing area in which cuvettes wash occur

3 Switch on - off

(See also User Manual, chapter “Start up”)

3.1 Switching the system on

1. Ensure that the cover and the flaps are closed.
2. Ensure that the analyzer main switch is on the “ON” position:



3. If a printer is connected, switch it on manually.
4. Check the soft power button and press it when the light stops blinking:



The system starts the operating system and the **LIAISON[®] XS** software on the integrated Panel-PC.

3. Switch on - off

5. After system start, the LIAISON[®] XS software shows the Startup page:



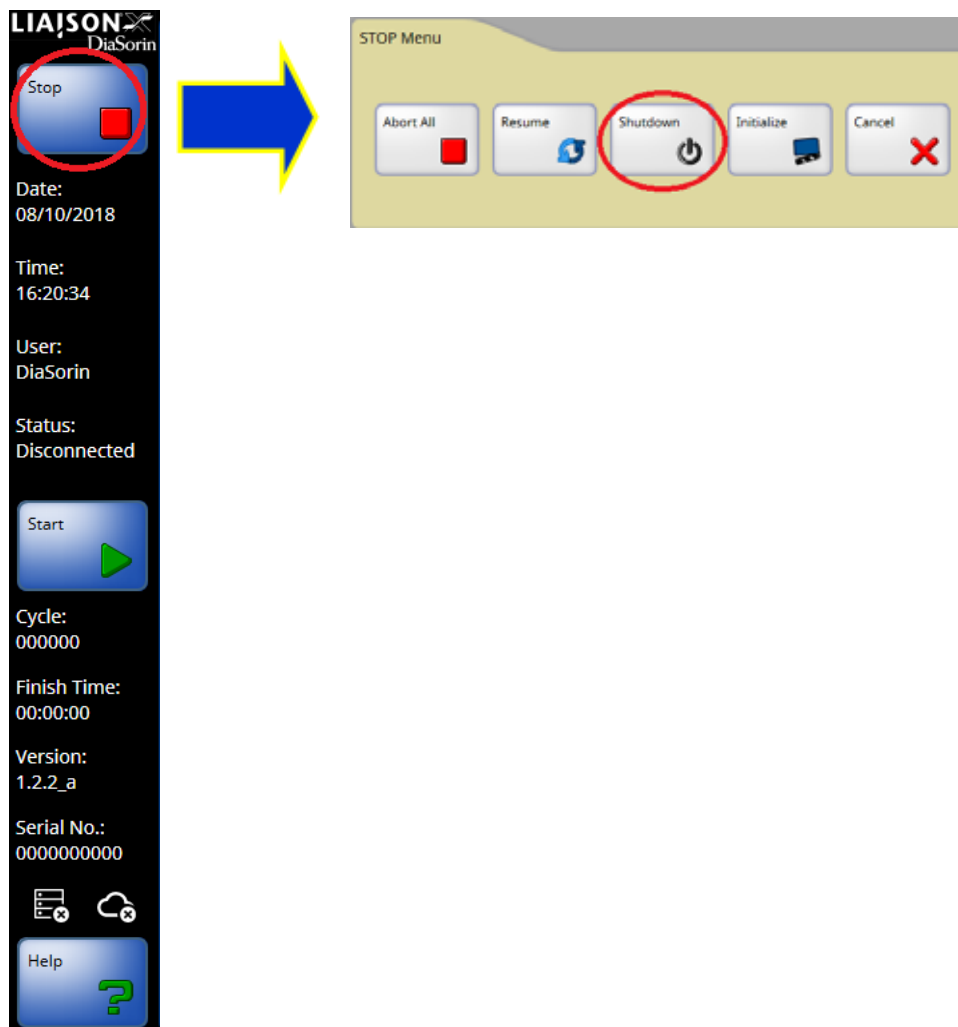
6. Press the "LiaisonXS" button. The Login page appears:



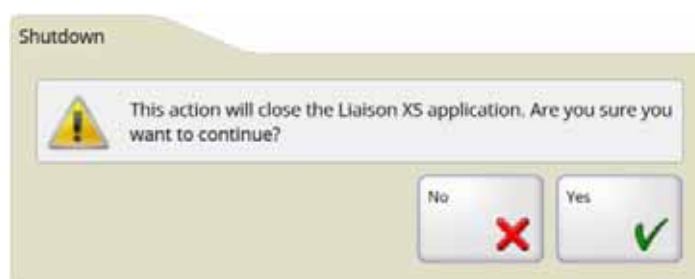
7. Type valid user name and password.
8. From the STOP menu, initialize the system.
9. After the initialization is completed, the system is in "Standby" and can be used for daily work as soon as the conditioning time is passed (1 hour).

3.2 Switching the system off

1. From the left header, select "Stop" → "Shut down":



2. Confirm the request with "Yes":



3. The PC switches off completely.
4. If a printer is connected, switch it off manually.
5. Switch off the analyzer main switch.

4 Calibration

(See also User Manual, chapter “Integrals, Calibrators, Ancillary reagents”)

4.1 Ancillary loading

This section applies only to kits that require an ancillary reagent. Handle it according to the related Instruction For Use.

1. Uncap ancillary vial.
2. Ensure that the vial contains no bubbles.
3. After requesting permission via the “Withdraw” button (the corresponding software LED should be grey), unload the 4-position rack from the Reagent Area.

Note: if the related software LED is already off, e.g. if the ancillary is not in use, the rack is already withdrawn.

4. Load the ancillary in an empty position of the dedicated rack:



5. Load the rack into the dedicated lane and check both the presence of the corresponding integral lots onboard and that the remaining volume is sufficient for the calibrations:



4.2 Integral loading

1. Remove the seals from integral vials (if present).
2. Ensure that neither foreign material nor liquid is present on the vial membranes and that the vials contain no bubbles.
3. Ensure that the compatible assay protocol is loaded on board.
4. Load the integral in the Reagent Area, handling it according to the related Instruction For Use (including loading into the integrated re-suspension tool, and leaving the integral to agitate for at least 15 minutes):

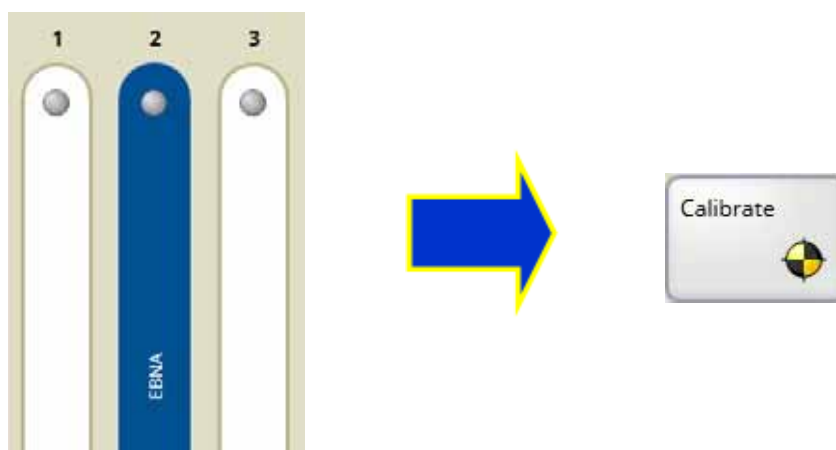


4. Calibration

5. In the case of internal calibrators, check that the number of remaining calibrations is greater than zero:

Original Number of Tests:	100
Remaining Tests:	85
Booked Tests:	0
Available Tests:	85
Remaining Calibrations:	2

6. In the case of external calibrators, ensure that they are loaded and recognized in the Sample Area.
7. A calibration can be created by selecting an integral from “Loading” → “Reagents”, then choosing “Calibrate”:



Note: the symbol  means that a calibration is “Created” for that integral

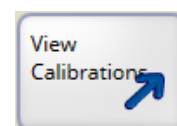
8. The calibration can be started pressing the “Start” button:



Note: the symbol  means that a calibration was started for that integral.

9. A calibration cannot be started if any of the following occurs:
- another calibration is in status “Created” or “Ongoing” for the same integral
 - another calibration is in status “Created” or “Ongoing” for another integral of the same lot (if that assay shares the Working Curve)
 - the calibrators are external and not present on board
 - any required resource is detected as missing or empty; see chapter 5.7 to check whether the available resources are sufficient.
10. When the calibration is completed, it will automatically be accepted or rejected.
11. The symbols  means that no valid calibration is present for that integral.
12. No specific symbol is shown if the selected integral has a valid calibration.

13. A calibration report can be seen by choosing “View Calibration” for a selected integral, then pressing the button “Details”:




Note: the same page can be reached by selecting “Results” → “Calibrations”.

Important notes:

- All calibrations (Valid, Failed, Expired,...) are reported in the page “Results” → “Calibrations”
- A sample test can be performed without a valid calibration: in this case, only RLU results will be given
- As soon as a valid calibration is performed, the system will automatically assign a dose to all sample results, performed within the last 18 hours, that have a valid RLU result and not a dose
- A calibration failure does not invalidate a previously existing valid calibration.

5 Test performance

(See also User Manual, chapter “Load Patient Samples or Controls and Assign Assays”)

5.1 Sample Loading

1. Check if the sample tubes are uncapped. Check that the sample volume is sufficient to run the required amount of test (as described in the Instruction For Use of the kits being used).
2. In the case of plasma gel separator containers, the amount of sample should be at least 500 µL more than the volume required to run the test.
3. Check for and remove air bubbles. Ensure no clots are present.
4. Load samples in appropriate racks, ensuring that the sample tubes fit with the chosen rack type:

Rack type	Use	Rack type	Use
B	15 mm (internal diameter)	X	10 mm (internal diameter)
W	14 mm (internal diameter)	G	Customizable tube
E	13 mm (internal diameter)	O	Customizable tube
F	12 mm (internal diameter)	Y	Customizable tube
I	11 mm (internal diameter)	Z	Customizable tube
A	10 mm (internal diameter)	C	Use only for DiaSorin special large vials (i.e. some types of control bottles or external calibrators)
J	9 mm (internal diameter)	H	LIAISON [®] Stool extraction device (X0034)
K	8 mm (internal diameter)	L	Used only for DiaSorin S.p.A. glass vials (i.e. some types of control bottles or external calibrators)
N	7 mm (internal diameter)	M	Calprotectin extraction device (X0043)
Q	6 mm (internal diameter)	P	Only for pediatric tubes

5. Ensure that the barcode (if present) is clearly visible through the side opening of the rack.

6. Load sample racks into Sample Area:



7. Manually type sample identifiers in case the barcode is not present.
8. Sample status is shown according to the following table:

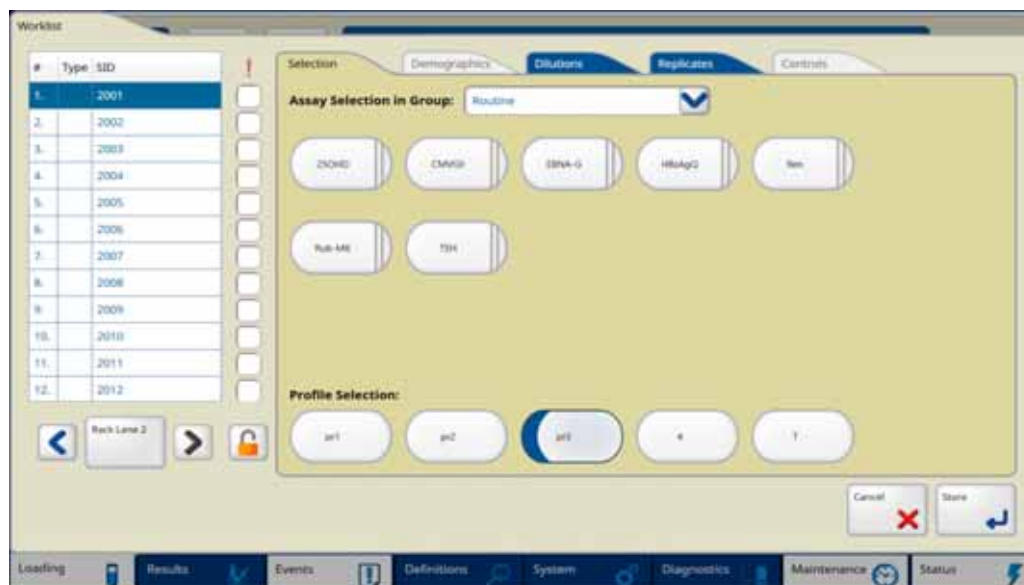
Symbol	Description
	Loaded sample tube with known SID .
	Loaded sample tube without bar-code or unreadable bar code, or no tube loaded. The SID is empty.
	Sample in process.
	Sample off-line, a clot detection or "No Liquid Found" occurred since last loading. (If a sample is off-line, unload it, check its status and reload again when OK.)
	A sample error occurred since last loading and at least one scheduled aspiration has to be completed.
	A sample error occurred since last loading and no scheduled aspiration has to be completed.

9. Special sample types are external calibrators  , controls  and light check  .
10. High-priority samples (STAT) are marked as  .

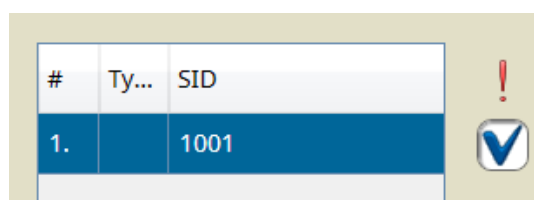
5. Test performance

5.2 Assay assignment for patient sample

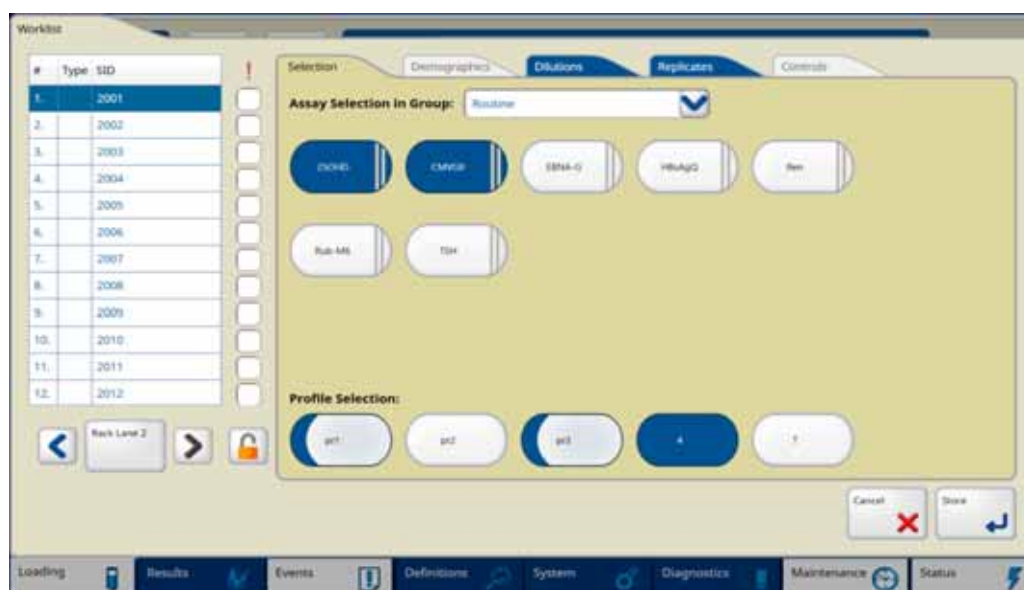
1. Select a rack and press “Schedule”, the following page will open:



2. Select one or more patient samples (clicking on the “Rack lane” button toggles between “Select All” and “Unselect All”). Include patient samples only. Use the  button to enable/disable multiple selection.
3. The following possibilities are available:
 - check the checkbox to declare the sample as high priority (STAT):



- from “Selection”, assign one or more assays or profiles to samples



Note: as soon as a test is started, it gets unselected and may be assigned again, even if the previous test is still running

The following table shows how assays are displayed:

Style	Description
	Integral currently not loaded on the instrument Assay not assigned to the selected sample
	Integral currently not loaded on the instrument Assay assigned to at least one selected sample
	Integral currently not loaded on the instrument Assay assigned to the selected sample
	Integral currently loaded on the instrument Assay not assigned to the selected sample
	Integral currently loaded on the instrument Assay assigned to at least one selected sample
	Integral currently loaded on the instrument Assay assigned to the selected sample

5. Test performance

- from “Demographics”, specify patient personal information

Worklist

#	Type	SID
1.		2001
2.		2002
3.		2003
4.		2004
5.		2005
6.		2006
7.		2007
8.		2008
9.		2009
10.		2010
11.		2011
12.		2012

Selection Demographics Dilutions Replicates Controls

Sample ID: 2001

Patient ID:

First name:

Initials:

Last name:

Date of birth:

Patient location:

Sender:

Sex: Male ☐ Female ☐ Unknown ☒

Back Lane 2 Lock

Cancel Store

Loading Results Events Definitions System Diagnostics Maintenance Status

- from “Dilutions”, assign one or more dilution factors to assigned assays (“Undil.” for undiluted tests)

Worklist

#	Type	SID
1.		2001
2.		2002
3.		2003
4.		2004
5.		2005
6.		2006
7.		2007
8.		2008
9.		2009
10.		2010
11.		2011
12.		2012

Selection Demographics Dilutions Replicates Controls

Selected Assay(s): 2

Available Dilutions:

S-X: 20040

F-X: V1

Undiluted

1:10

1:100

1:200

Back Lane 2 Lock

Cancel Store

Loading Results Events Definitions System Diagnostics Maintenance Status

- from “Replicates”, change the number of replicates for assigned assays

#	Type	SID
1.		LITTEHEN
2.		HAUSER
3.		SAMPLE031
4.		SAMPLE039
5.		DONNER
6.		HAMPELMANN
7.		TOOS2
8.		SAMPLE037
9.		SAMPLE035
10.		HANSTVOGEL
11.		SAMPLE029
12.		SAMPLE030

Note: only a number of replicates equal to or higher than the default value can be selected

- After completion, press “Store” to confirm and close the page, or choose the next rack using  .

5.3 Assay assignment for control samples

- Select a rack and press “Schedule”, the following page will open:

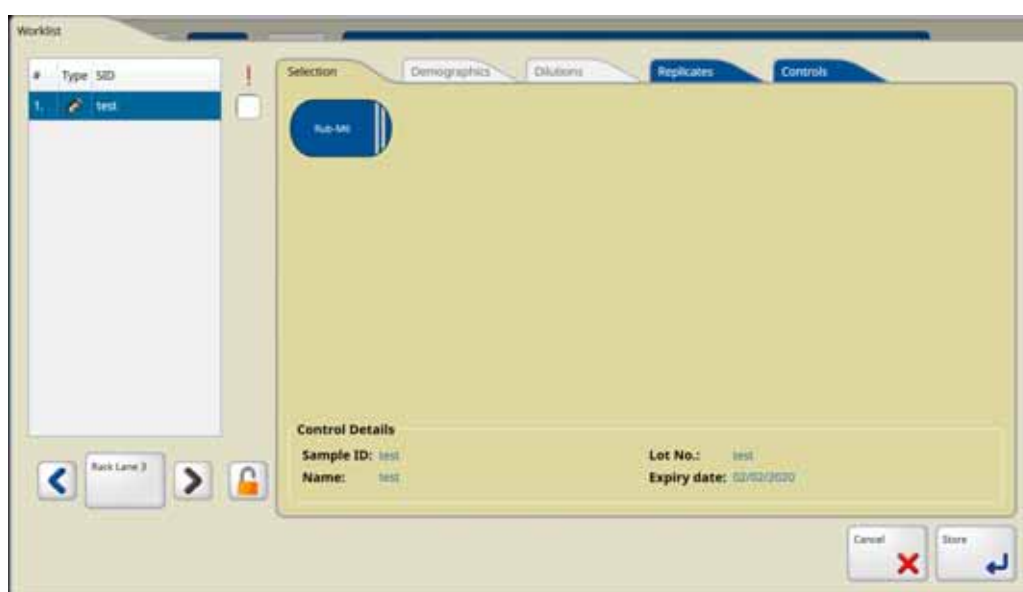
#	Type	SID
1.		test

5. Test performance

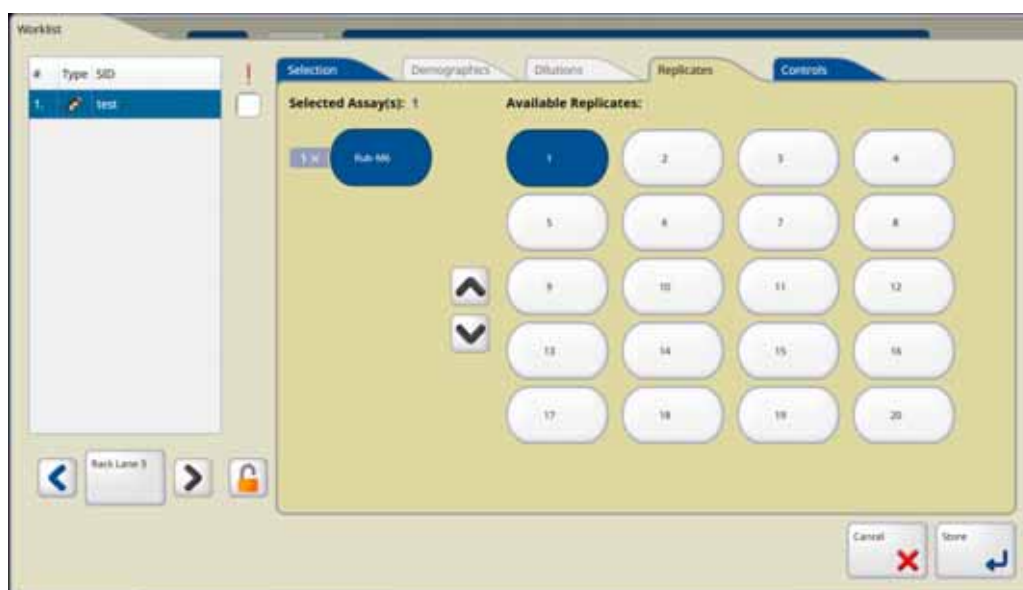
2. Select one control sample.
3. The following possibilities are available:
 - check the checkbox to declare the control as high priority (STAT):

#	Ty...	SID
1.		1001
2.		1234

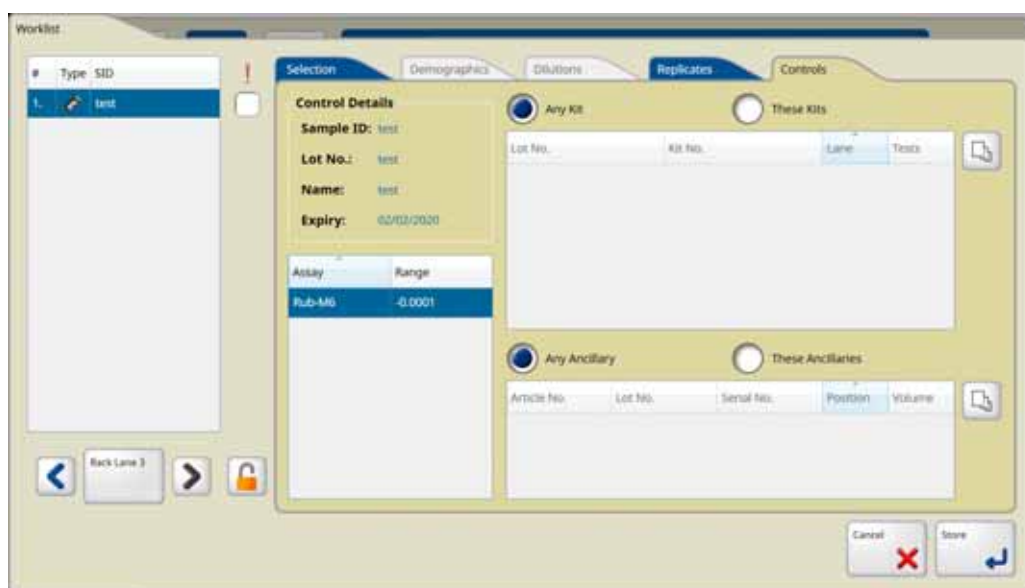
- from “Selection”, assign one or more assays to controls:



- from “Replicates”, change the number of replicates for assigned assays



- from “Controls”, check the loaded integrals/ancillaries for an assay and choose which one to use:



Note: as soon as a test is started, it gets unselected and may be assigned again, even if the previous test is still running.

The following table shows how assays are displayed:

Style	Description
	Integral currently not loaded on the instrument Assay not assigned to the selected sample
	Integral currently not loaded on the instrument Assay assigned to the selected sample
	Integral currently loaded on the instrument Assay not assigned to the selected sample
	Integral currently loaded on the instrument Assay assigned to the selected sample

- After completion, press “Store” to confirm and close the page, or choose the next rack using .

5.4 Assay assignment for external calibrators

- There is no assignment available for external calibrators.

5.5 Integrals loading

1. Remove the seals from integral vials (if present).
2. Ensure that neither foreign material nor liquid is present on the vial membranes and that the vials contain no bubbles.
3. Ensure that the compatible assay protocol is loaded on board.
4. Load the integral into the Reagent Area, handling it according to the related Instruction For Use (including loading into the integrated re-suspension tool, and leaving the integral to agitate for at least 15 minutes).
5. Check that the number of available determinations for calibrated integrals is sufficient for the incoming routine:



5.6 Ancillary loading

1. This section applies only to kits that require an ancillary reagent. Handle it according to the related Instruction For Use.
2. Uncap ancillary vial.
3. Ensure that the vial contains no bubbles.
4. After asking for permission via the "Withdraw" button (the corresponding software LED shall be grey), unload the 4-positions rack from the Reagent Area.

Note: if the related software LED's are already off, e.g. if the ancillary is not in use, the rack is already withdrawn.

5. Load the ancillary into an empty position of the dedicated rack.

- Load the rack into the dedicated lane, check both the presence of corresponding integral lots and that the remaining volume is sufficient for the upcoming routine:



5.7 Upfront estimation

- To check whether any resource might not be sufficient to complete the requested tests, select "Status" → "Needs" and press "Calculate":

Search:

Last Update: 04/30/2020 13:36:16 Total Records: 9

Name	Placed	Origing	Sum	Not Booked
QFT	3	0	3	28
M0101	0.036ml	0.000ml	0.036ml	88.885ml
Cuvettes	3	0	3	142
SolidWaste	3	0	3	167
Tips	3	0	3	187
Starters	3	0	3	881
LiquidWaste	79ml	0ml	79ml	N/A
System Liquid	54ml	0ml	54ml	N/A
WashBuffer	24ml	0ml	24ml	N/A

Calculate

Summary Plastics Needs Inventory

Loading Results Events Definitions System Diagnostics Maintenance Status

Note: check the value in "Not Booked" column. If positive, the resources are available to process the requested tests; if negative, there are not enough resources to complete the requested tests.

For System Liquid, Wash Buffer and Liquid Waste visually check if the level of the tanks is compatible with the needed volume displayed in the "Sum" column.

5.8 Routine start

The programmed calibrators, controls and patients tests will start by pressing the “Start” button.

6 Results

(See also User Manual, chapter “Main Category Results”)

1. Selecting “Results”, the following pages will be available for the 2nd level choice:

Page	Type	Content
All	Independent result set	All tests with and without results, that have neither been archived nor deleted
Worklist	Subset of “All”	All programmed tests (in “To Do” and “Placed” status)
Ongoing	Subset of “All”	All running and not completed tests (in “Scheduled”, “Active” and “Measured” status)
Done	Subset of “All”	All successfully completed tests (in “Done” status)
Archived	Independent result set	All tests that have been archived
Failed	Subset of “All”	All unsuccessfully completed tests (in “Failed” status)
Calibrations	Independent calibration set	All calibration reports (in any status)
Controls	Subset of “All”	All control results with and without results, that have neither been archived nor deleted
History	Temporary result set	History results and results gathered from “All” and “Archived” pages by an explicit user query.

Note: deleting or archiving a test result in “All” will have the same effect in all its subsets, and vice versa.

Search: Total Records: 49

Sample ID	Assay	Measured	Status	Dilution	BLI	Flags
T0057	ESNA-G	---	Placed	1		
SAMPLE037	ESNA-G	---	Placed	1		
SAMPLE035	ESNA-G	---	Placed	1		
HANSEVOGEL	ESNA-G	---	Placed	1		
SAMPLE029	ESNA-G	---	Placed	1		
SAMPLE030	ESNA-G	---	Placed	1		
00005	LogicFam	16.08.2018	Done	1		
00001	LogicFam	16.08.2018	Done	1		
00003	LogicFam	14.08.2018	Done	1		
00002	LogicFam	14.08.2018	Done	1		
00001	LogicFam	14.08.2018	Done	1		
ISonL38B	SonL3	17.08.2018	Measured	1	763	
ISonL38A	SonL3	17.08.2018	Measured	1	1551	

Buttons: Delete, Export, Download, Filter, Print, Details

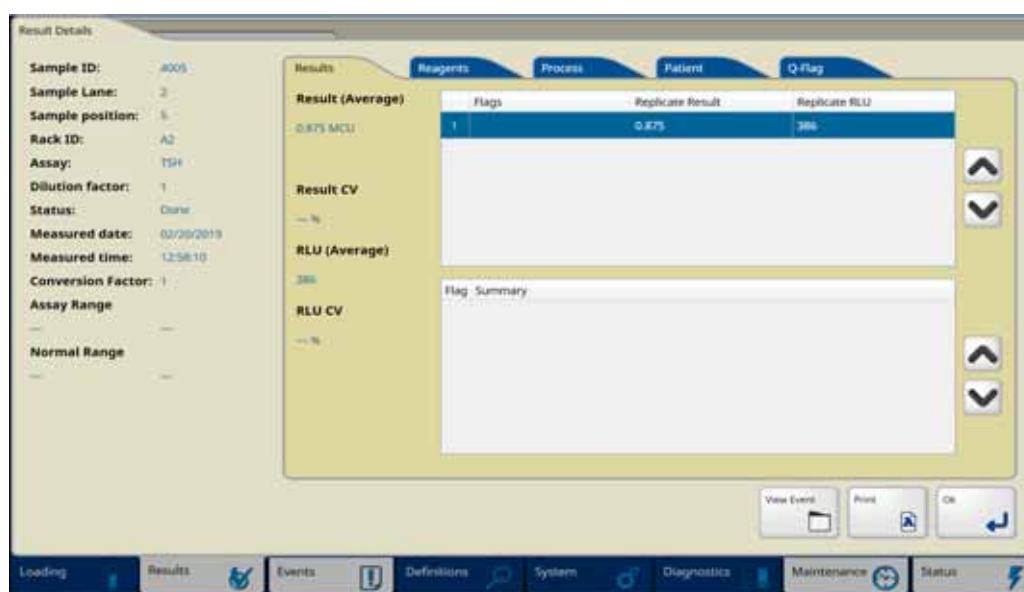
Navigation: All, Worklist, Ongoing, Done, Archived, Failed, Calibrations, Controls, History

Bottom Bar: Loading, Results, Events, Definitions, System, Diagnostics, Maintenance, Status

6. Results

2. Each test result (excluding calibrations) is comprised of the following data:

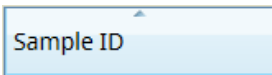
Data	Meaning	Examples
General information	Test result main characteristic	Sample ID, Assay, Rack lane
Result	Raw and elaborated data, as outcome of the immunometrical analysis	RLU, Dose, Qualitative label, Flags
Reagents	Traceability information about the used reagents	Kit lot No., Calibration ID
Process	Traceability information about the occurred analytical and post-analytical process	Processing Time, Sent to LIS by
Patient	Traceability information about the patient related to the performed test (only for patient samples)	Name, Date of Birth
Qflag	List of Run Validation Criteria (RVC) and control not satisfied (only for patient samples)	Control name, Kit, Calibration



3. The status of each test can be one out of the following table:

Status	Meaning
To Do	Sample not present on board, not started yet
Placed	Sample present on board, not started yet
Scheduled	Starting soon
Active	Under performance
Measured	Completed but no dose received
Done	Successfully completed
Failed	Unsuccessfully completed

4. The following table shows the operations available in the various pages:

Button	Meaning	Page
	Sorts the contents of the table according to the items of the selected column (if the column does not support sorting, the small triangle will not appear upon clicking). Note: special characters are ignored.	All pages
	Searches the first item that begins with the typed characters, looking into the same column currently used for sorting. Note: not case sensitive, special characters are ignored.	All pages
	Scroll the text upwards and downwards, respectively: - all the way up/down, - item by item up/down - page by page up/down	All pages
	Enable/disable multiple selection	All pages
	Opens the "Result Details" screen for the selected entry, to check all result details	All pages
	Prints a summary report of all selected results	All pages
	Opens the "Filter Definition" dialog, to apply a filter criterion on the page results	All pages
	Deletes the selected results	All pages except "Ongoing"
	Queries LIS for tests to be performed	"All", "Worklist"
	Exports the selected results in text format to a file	"All", "Done", "Archived", "Calibrations", "Controls", "History"
	Turns the test into "failed" and aborts the related processing	"Ongoing"
	Archives the currently selected entries, i.e. moves them to the "Archived" page	"Done", "Failed"
	Opens the "Rerun" dialog, to create a new test with the same characteristics as the selected one	"Done", "Failed"

6. Results

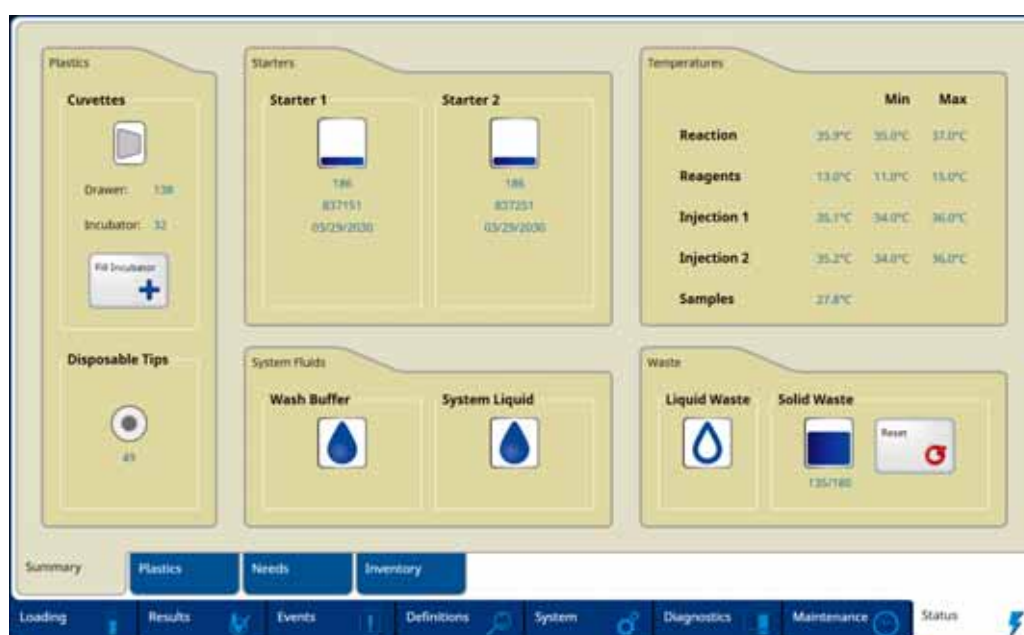
	Transmits the selected results to the LIS	"Archived"
	Opens the "Reagent loading" page to look at loaded reagent status	"Calibrations"
	Recalculates the currently selected entries, using the most recent calibration for that assay. For Combi Assays, all results for the same assay family are recalculated. Do not use for "Active" tests.	"Ongoing", "Done", "Failed"
	Makes all the selected "Valid" and "Created" calibrations not usable anymore.	"Calibrations"
	Opens the "Query" dialog to apply a query on both historicized results and results gathered from "All" and "Archived" pages	"History"

7 Consumables & Temperatures

(See also User Manual, chapters “Check Cuvettes, Disposable Tips, Starter Reagents and Liquid Containers/Tanks”, “Main Category Status” and Consumables Instruction For Use)

7.1 Consumables value check

- To check the status of consumables, select “Status” → “Summary”.



Note: the status of all consumables is continuously monitored.

- The following table contains the available status for each resource:

Consumable	Red phase	Other status	Additional information
System Liquid	1.8 l	Absent, Not primed	System liquid is also used for periodical SPOLV rinsing in Standby or Ready status
Wash Buffer	1.8 l	Absent, Not Primed	-
Liquid Waste	7 l	Absent	-

7. Consumables & Temperatures

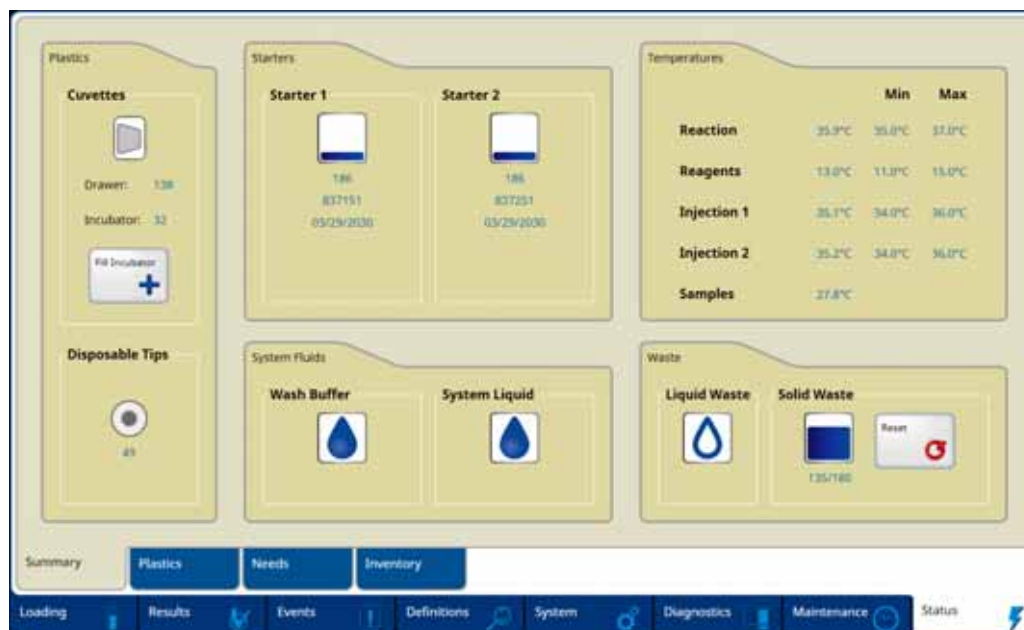
Consumable	Availability threshold	Other status	Additional information
Starter Reagents	4	Absent, Not Primed, Expired, Error	Starters will report as depleted even if they have more than 4 shots left, if the system doesn't have sufficient shots to prime them
Cuvettes	2	Absent (drawer pulled out)	The threshold for cuvettes refers to the number of cuvettes in the trays
Solid Waste	178	Absent (pulled out)	-
Tips	0	Absent (drawer pulled out)	-

3. In the case a tank enters red phase, the system will continue tests that have already been started (i.e. the ones in status "Active"), but will deny starting any new ones and will turn the "Scheduled" jobs in "Placed".
4. The "Status" toggle will start flashing in the following situations:

Consumable	Conditions
Starter Reagents	Below availability threshold or in status "Absent", "Expired", "Error"; also "Not Primed" if the system is in Ready status
Cuvettes	Below availability threshold or "Absent"
Solid Waste	Above availability threshold or "Absent"
Tips	Below availability threshold or "Absent"
System Liquid	Below red phase or "Absent"; also "Not Primed" if the system is in Ready status
Wash Buffer	Below red phase or "Absent"; also "Not Primed" if the system is in Ready status
Liquid Waste	Above red phase or "Absent"

7.2 Temperature value check

- To check the status of temperatures, select “Status” → “Summary”:



- The reported status color reflects the adequateness of each temperature, according to the specific acceptance range, reported next to the measured value, like in the examples below:

Displayed				Meaning
		Min	Max	
Reaction	36.0°C	26.0°C	46.0°C	The temperature is within the acceptance ranges
Injection 1	33.2°C	34.0°C	36.0°C	The temperature exceeds the acceptance ranges

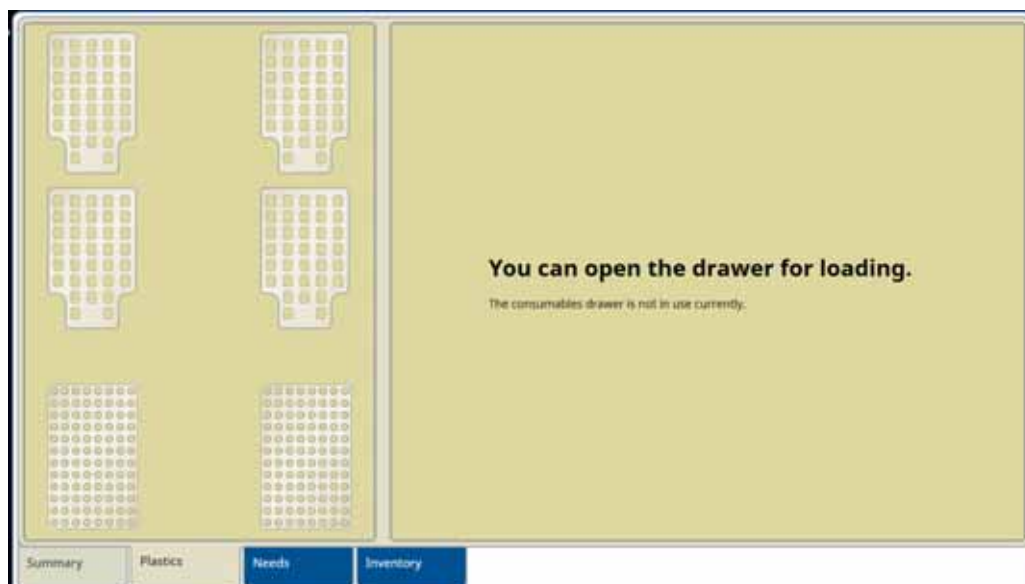
Note: The “Samples” temperature has no acceptance range.

- In case a temperature is persistently shown as **red** color, please contact the local technical support.

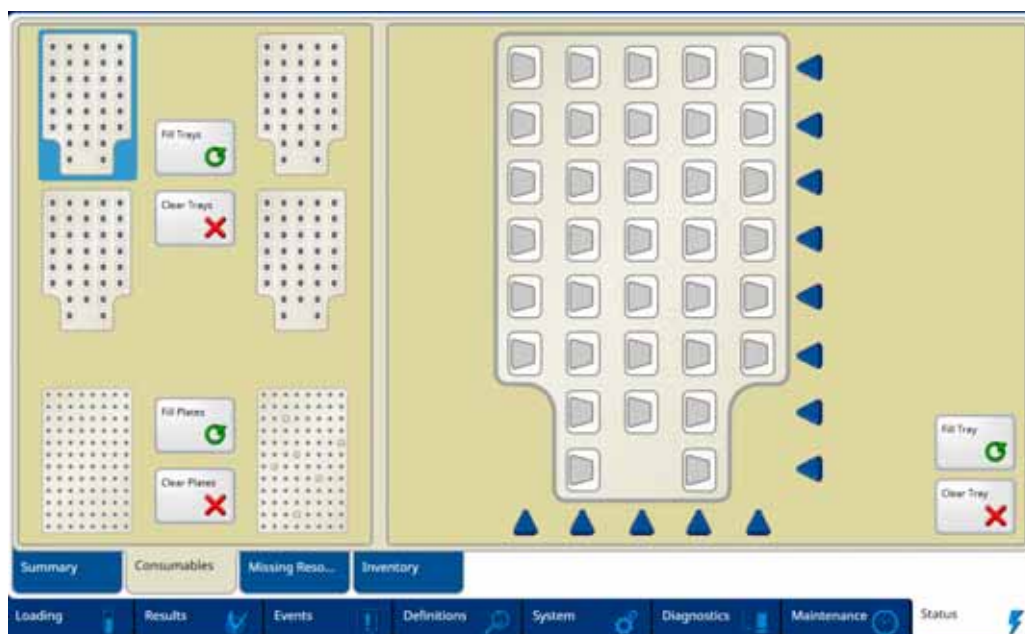
7.3 Cuvette & Tip Loading

7.3.1 Plastics drawer not in use

1. Select the page “Status” → “Plastics”:

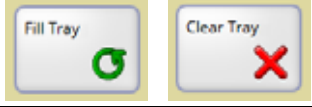
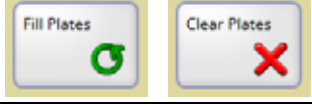
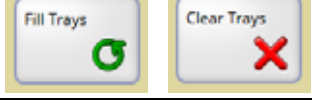


2. Open the drawer, the following is displayed:



3. Replace cuvettes and tips trays with new ones

4. Update the plastics content of each tray acting on:

each single tip position	
each single cuvette position	
each column/row	
the buttons "Fill/Clear Plate"	
the buttons "Fill/Clear Tray"	
the buttons "Fill/Clear Plates"	
the buttons "Fill/Clear Trays"	

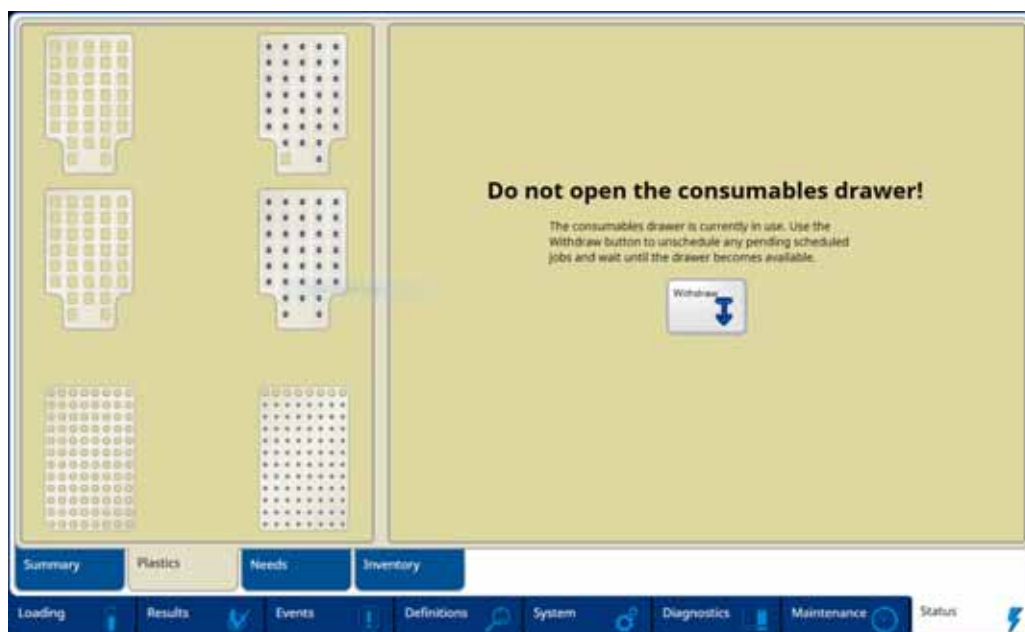
5. Close the drawer to reassign it to the system

7. Consumables & Temperatures

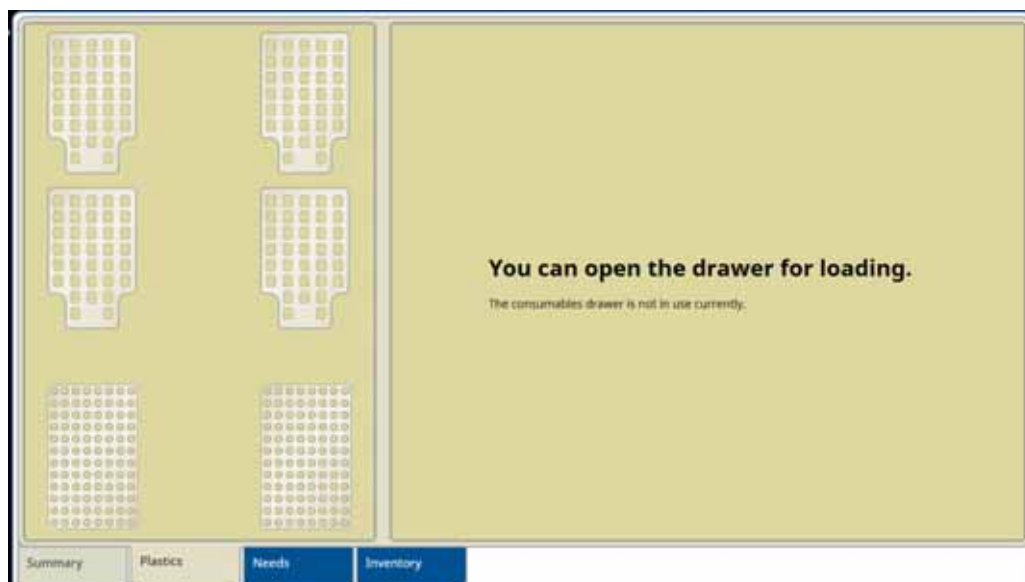
7.3.2 Plastics drawer in use

When the system is accessing the drawer, it is possible to suspend that access: the jobs in “Scheduled” status will return in “Placed” status.

1. Select the page “Status” → “Plastics”:



2. Click on the button “Withdraw”
3. Wait until the countdown ends and the following is displayed:



4. Open the drawer and refill the plastics as described in chapter 7.3.1.
5. Close the drawer to reassign it to the system.

Note: if jobs were returned to “Placed”, it is necessary to press “Start” to process them.

7.4 Other consumables loading

1. Starter reagent bottles, System Liquid and Wash Buffer can be unloaded and reloaded when the system is in “Standby” or “Ready” status. If they are removed while the system is using them (“Running”, “Maintenance”), the system will abort any ongoing test/task and unschedule any scheduled test.
2. Liquid Waste can be unloaded and loaded when the system is in “Standby” or “Ready” status. If it is removed while the system is using it (“Initialization”, “Running”, “Maintenance”), the system issues a fatal error and moves to “Halted” status.
3. Solid Waste Drawer can be opened and closed when the system is in “Standby” or “Ready” status. If it is opened while the system is using it (“Initialization”, “Running”, “Maintenance”), the system issues a fatal error and moves to “Halted” status. In case the Easy Waste box has been replaced with a new empty one,

press the button “Reset”



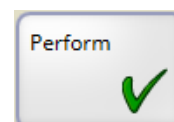
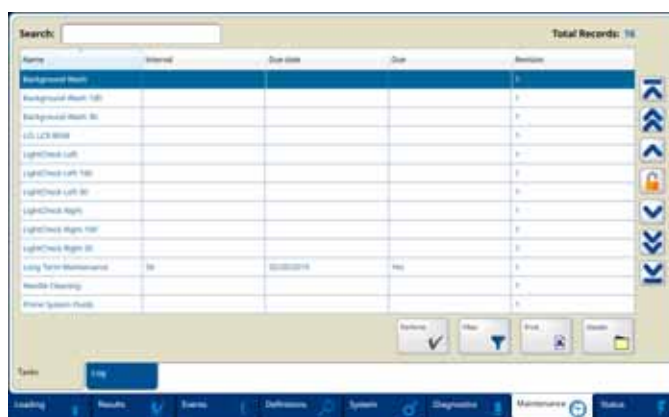
in order to reset the solid waste counter.

8 Maintenance & back-up

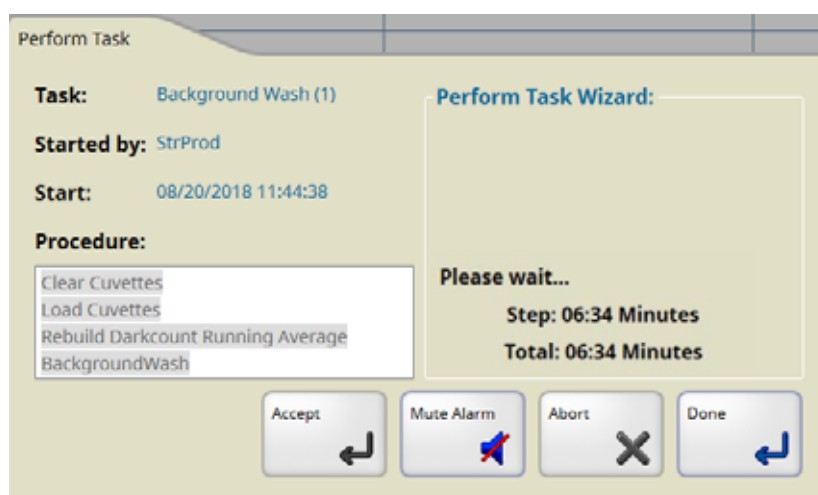
(See also User Manual, chapters “Main Category Maintenance” and “Main Category System”)

8.1 Maintenance

1. The maintenance is scheduled on a regular basis. Whenever a maintenance task is due, the “Maintenance” tab starts flashing. If so, or even before the due time, it is necessary to perform the maintenance task.
2. To perform a maintenance task, ensure that the system is in “Stand-by” or “Ready” status, then select “Maintenance” → “Task”, check which task is due, select it and press “Perform”:

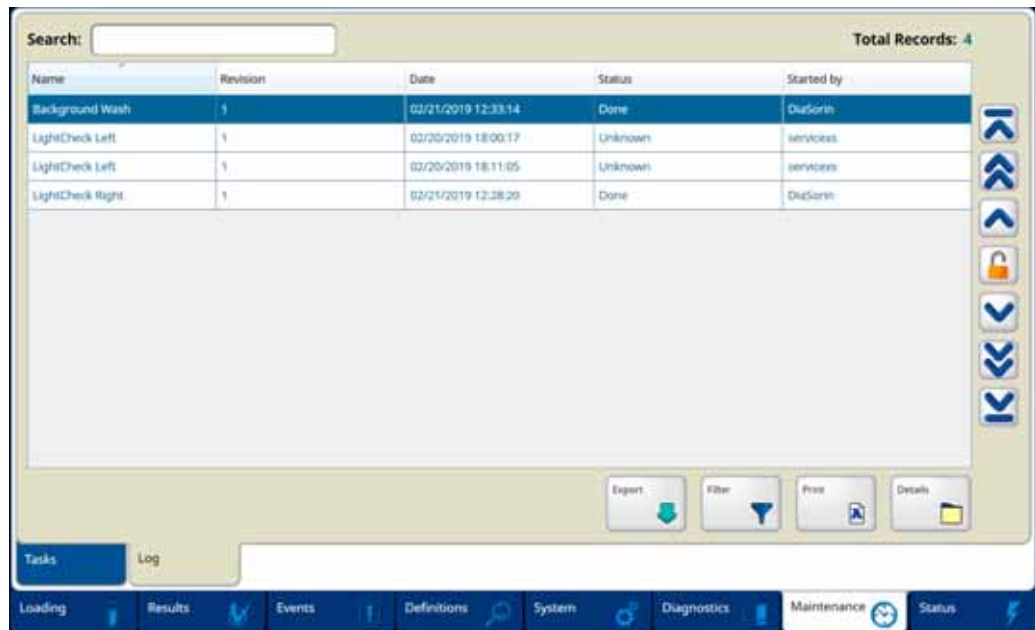


3. The dialog “Perform Task” will open; follow its indications up to the end:



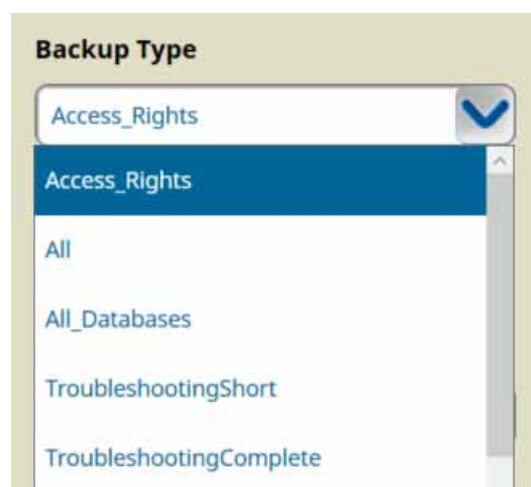
Note: during a Maintenance task it will not be possible to do anything else on the system.

- It is possible to check the results of the performed maintenance tasks: select "Maintenance" → "Log" to view, print or export them to a file:



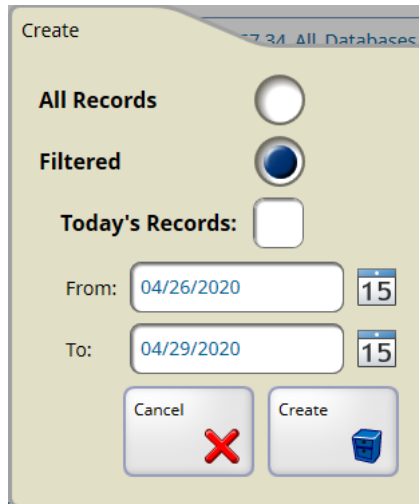
8.2 Back-up

- It is possible to save and export to an USB stick or a network path any of the pre-configured set of data.
Note: the system makes automatic data back-ups for data safety purposes; this data can be retrieved as well, following the same procedures here described.
- Select "System" → "Backup" and ensure that the system is not running. It will not be possible to start new jobs until the procedure is completed.
- To create a new back-up, select an item from the pick-up list "Backup Type" and press "Create":



8. Maintenance & back-up

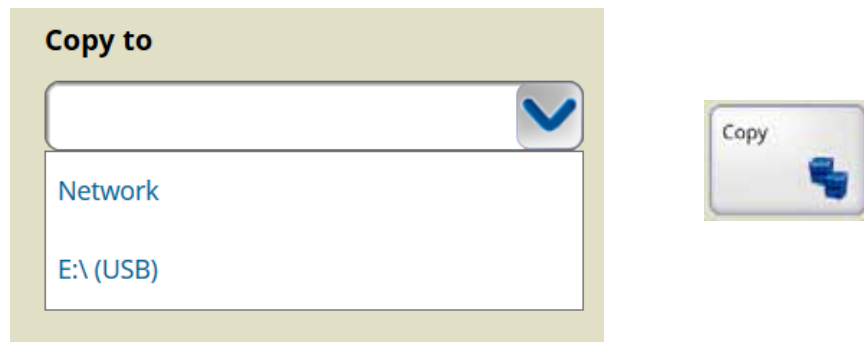
4. Upon selection of an applicable backup type, a dialog is opened to allow the user to apply a time based filter. The selected backup type will include files on the basis of the selected period:



The 'Create' dialog box is shown with the following options:

- All Records**: Unselected (radio button).
- Filtered**: Selected (radio button).
- Today's Records**: Unselected (checkbox).
- From**: 04/26/2020 (calendar icon shows 15).
- To**: 04/29/2020 (calendar icon shows 15).
- Buttons**: 'Cancel' (with a red X icon) and 'Create' (with a blue cube icon).

5. To copy an already existing back-up to an USB stick or a network path, select a destination from "Select a Backup to Copy", select one existing item from the related table and press "Copy":



The 'Copy to' dialog box shows a list of destinations:

- Network
- E:\ (USB)

To the right of the dialog box is a 'Copy' button with a blue cube icon.

Note: the network path has to be configured by the Field Service Engineer

6. After the procedure is completed, it is possible to start new routines.

9 Assay definition

(See also User Manual, chapter “Main Category Definitions”)

1. To check or modify Assay parameters, or to import a new or updated Assay definition, select “Definitions” → “Assay”:

Search: Total Records: 13

Assay	LIS Alias	Revision	Article No.	Group
AFP	AFP	A2	314471	
EBNA-G	EBNA-G	VU01	310520	ALL
LogicFam	LogicFam	A2	555555	Combi
QantFam	QantFam	A2	889999	Combi
RUB-FULL	RUB-FULL	A1	310721	
SonL3	SonL3	A1	303300	Combi
SonL4	SonL4	A1	404400	Combi
SonQ5	SonQ5	A2	55555	Combi
SonQ6	SonQ6	A2	66666	Combi
SonQ7	SonQ7	A2	77777	Combi
SonQ8	SonQ8	A2	88888	Combi
Test1	Test1	VU01	666666	
Test2	Test2	A1	3185987	

Buttons: Import, Print, Details

Navigation: Assays, Controls, Groups, Run Matrix, Profiles, Senders

Bottom Bar: Loading, Results, Events, Definitions, System, Diagnostics, Maintenance, Status

2. To import a new or updated Assay definition, press “Import” and select the location where the assay protocol has been previously saved.
Note: if the Assay is new, it must be included in a Group to use it (“Definitions” → “Groups”).
3. To check or modify an Assay parameter, press “Details”:

25OHVD Details

Name: 25OHVD

LIS Alias: 25OHVD

Abbreviation: 25OHVD

Article No.: 310600

Creation Date: 08/28/2016

Revision: A1

On Board Stability: 28

Sample Replicates: 1

Control Replicates: 1

Priority: 3

Groups: A2

Calculation: Calibration Ranges Dilutions RVC Audit Trails

Default Unit: ng/ml

User Unit:

Conversion Factor:

Digits: 3

Send Replicates to Host: Yes ☐ No ☒

Result Aspect for Host Transmission: Dose ☒ Label ☐ Both ☐

Buttons: Cancel, OK

9. Assay definition

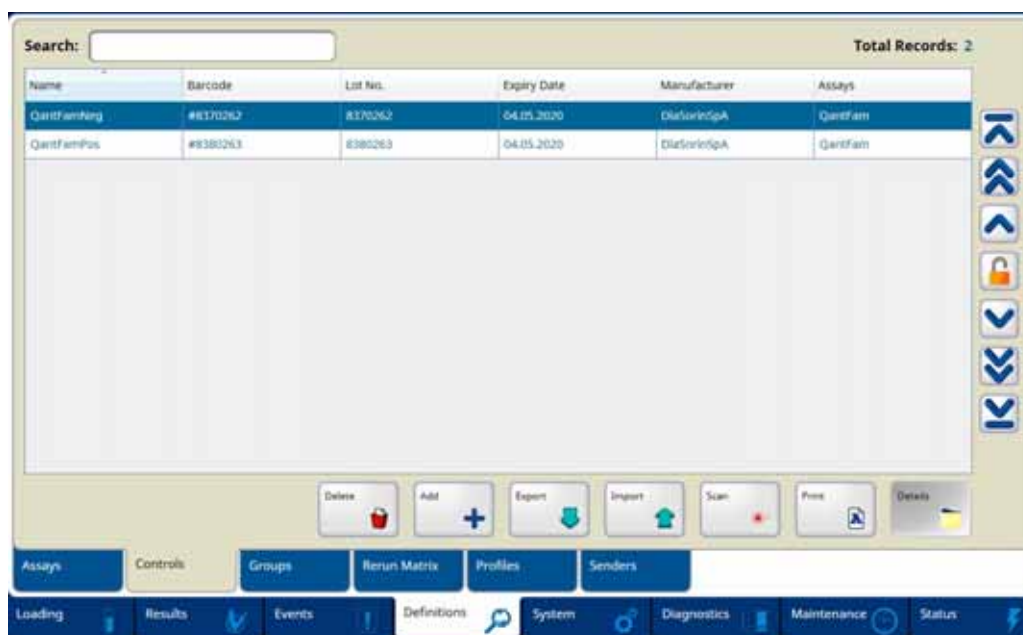
Note: the tab from “Calculation” to “Audit trails” can be toggled to get access to all available information for an Assay.

4. The fields that have a highlighted background are fields containing an incorrect value (e.g. it can be a mandatory field).
Note: importing a new version of an Assay definition will maintain the user’s changes and related Audit Trails, unless explained differently by the system.
5. After the changes, it is necessary to type a reason, which will be kept in the “Audit trail” tab, together with the user, time, original and final value.
6. When the check/change is complete, press “Cancel” to discard the changes or “OK” to keep them.
7. The “Print” button in the “Definitions”, “Assays” page allows to print a summary of all selected Assays.

10 Control definition

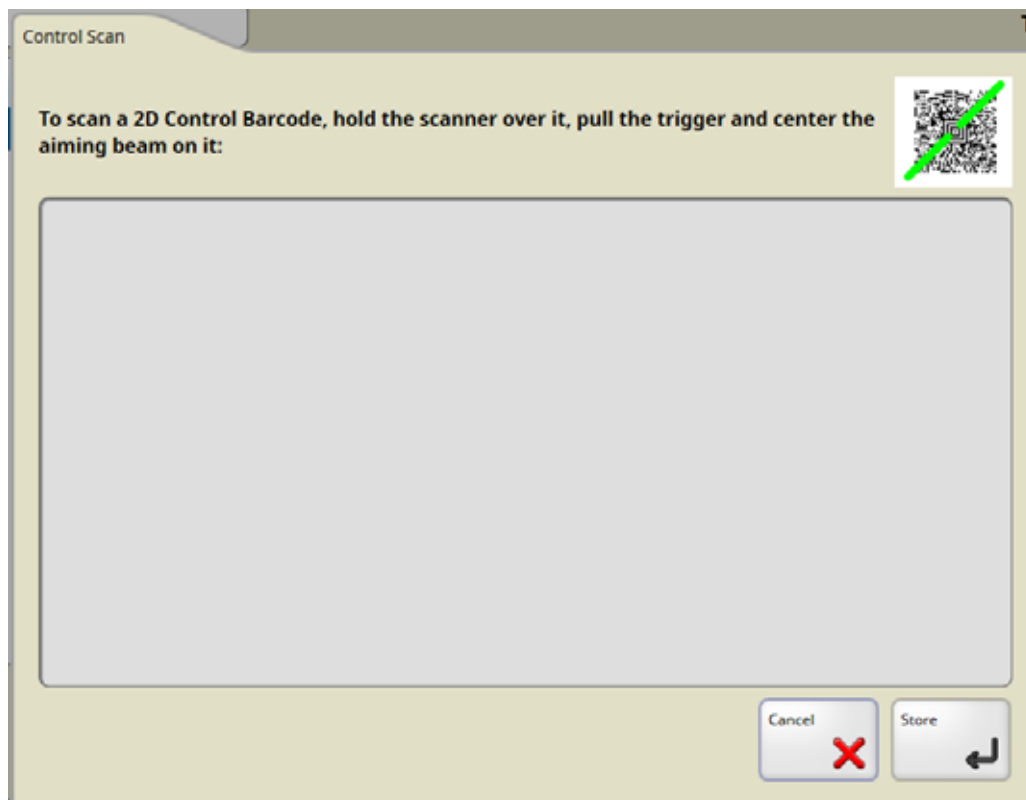
(See also User Manual, chapter “Main Category Definitions”)

1. To introduce, check or modify Control parameters, select “Definitions” → “Controls”:



10. Control definition

2. To introduce a new Control definition, for Controls provided with a dedicated 2 dimensional barcode, press "Scan" and follow the given instructions; then press "Store" to store all controls data:



3. To introduce a new Control definition, for Controls not provided with a dedicated 2 dimensional barcode:
 - press "Add" and fill the control details data,
 - press "Add", select an assay and press "Select" (the assay is added to the assay list on the left)

- select the assay in the list and fill the assay specific data on the right portion of the window

- repeat the previous two steps for all the involved assays
- press "Store" to store all controls data.

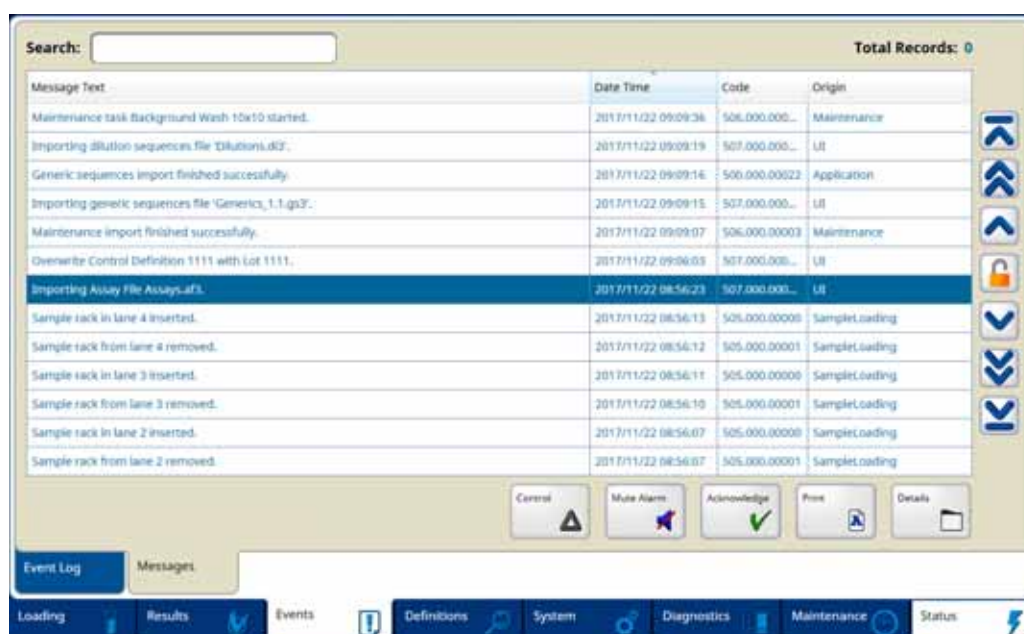
Note: the fields that have a grey surrounding can not be edited.

4. To move a Control definition from a LIAISON[®] XS PC to another, it's possible to select it and press "Export", choose a removable device, then move that device on a second PC, select "Import" and follow the directions.

11 Events

(See also User Manual, chapter “Main Category Events”)

- To ensure an adequate, real time information about what happened on the system, the “Events” pages are provided, featuring:
 - Event Log
 - Messages
- If a critical problem occurs on the system and the analyzer is not able to recover, a Fatal Error will be issued. It is necessary to initialize the system and, if the problem persists, contact the local technical support.
- If an event, that does not necessarily require the user’s immediate intervention but must be confirmed by the user, occurs on the system, it will be reported into the “Events”-“Messages” page and the related toggle will start flashing:



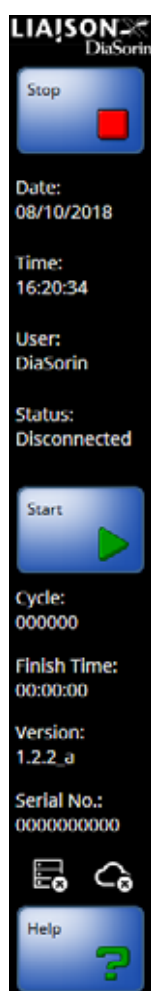
Note: select the items and press “Acknowledge” to confirm and remove messages from the page; their occurrence will remain reported in the “Events”, “Event Log” page. The “Mute Alarm” button will turn the alarm off.

- All the events that occurred on the system and that require adequate traceability will be reported into the “Events”-“Event Log” page.

12 Left Header

(See also User Manual, chapter “Left header”)

1. The header provides the following functionalities:



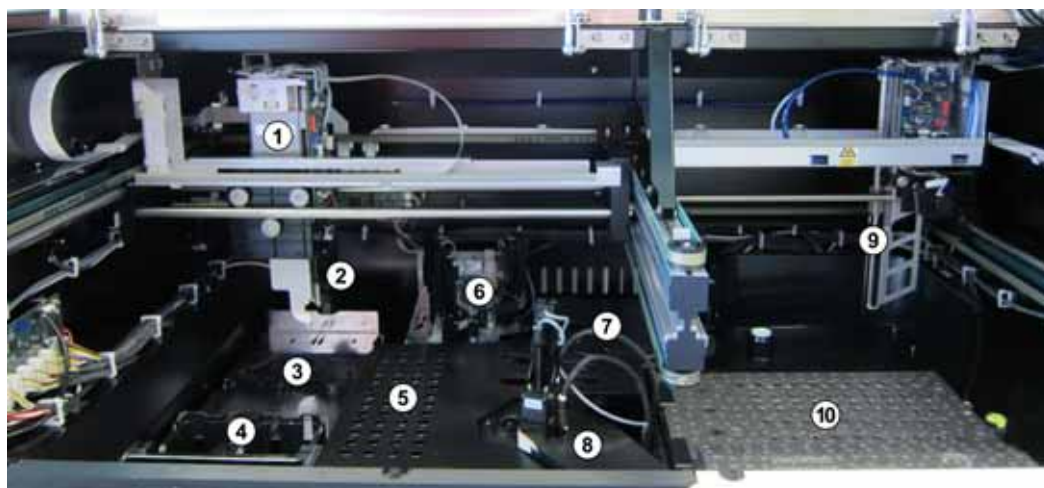
Button	Feature
	Allow to choose among the following actions: • Abort All: abort all ongoing test • Resume: start the resume • Shutdown: switch off PC and instrument • Initialize: initialize the system • Cancel: close with no action
	Start the tests in “Placed” status and, if in “Standby”, start the resume
	Open the Instruction For Use for reagents and instrument
	Indicator for LIS connectivity/activity
	Indicator for remote connectivity/activity

Annex 1: FLAGS

The flags are listed in order of priority.

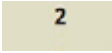
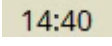
No.	Flag	Description
1	*	Mechanical error. Detailed information is available in the result details.
2	S	Error related to the pipetting with sample arm. Detailed information is available in the result details.
3	R	Error related to the pipetting with right arm. Detailed information is available in the result details.
4	C	Valid calibration missing (result without dose).
5	D	The sample was over-diluted. Only for Patient Samples diluted per user request.
6	Z	Mathematical error (division by zero) for Combi Assay calculation.
7	M	Mathematical error for dose calculation.
8	X>, X<, XM	For Combi results only, in case a Combi Son has a flag ">", "<" or "M" respectively.
9	E	Reagent onboard validity expired.
10	>, <	Result is above/below Assay Range or curve domain (for patient and control samples only).
11	NH, NL	Result is above/below Normal Range (for patient samples only).
12	QE	Control expired (for control samples only).
13	Q	At least one RVC was not satisfied. Controls (not defined as RVC) run before that patient sample have QH/QL flag. Based on last performance of any control defined for that assay. Only for patient samples.
14	QH, QL	Controls above/below the Manufacturer Range or outside the Assay Range, if a Manufacturer Range is defined (for control samples only).
15	UH, UL	Controls above/below the User Range or outside the Assay Range, if a Manufacturer Range is defined (for control samples only).
16	RM	The result was recalculated.
17	&	Result caused a rerun according to a Rerun Rule (for patient samples only).
18	X&, XNH, XHL	For Combi results only, in case a Combi Son has a flag "&", ">", "<", "NH" or "NL" respectively.

Annex 2: INSTRUMENT SCHEME (system modules)



No.	Description
1)	Air pipettor with disposable tip adapter for sample pipetting
2)	Cuvette gripper
3)	Cuvette trays area
4)	Disposable tips area
5)	Samples area
6)	Washer
7)	Incubator
8)	Reader
9)	Pipettor with steel needle (SPOLV) for reagent pipetting
10)	Reagents area

Annex 3: INTEGRAL SYMBOLS

		Symbol	Description
			Lane occupied
		 	Status: Not operative - may be removed Operative - Do not remove
			Assay name (if grey and in parenthesis: integral offline)
		     	Calibration status: No valid calibration + not started No valid calibration + created No valid calibration + started Valid calibration + not started Valid calibration + created Valid calibration + created
			Remaining determinations
			Agitation countdown